

EXPERIMENT-1

Chromatography

Aim: To separate the coloured components present in the mixture of red and blue inks by ascending paper chromatography and find their R_f values.

APPARATUS

Glass jar, glass rod, filter paper strip (Whatman No. 1 filter paper), jar cover and a fine capillary tube.

REQUIREMENTS

1. Take a Whatman filter paper strip (20x3cm) and draw a line with pencil above 3cm from one end. Draw another line lengthwise from the centre of the paper as shown in Fig. 6.1.
2. With the help of fine capillary tube, put a drop of the mixture of red and blue inks at the point P. Let it dry in air. Put another drop on the same spot and dry again. Repeat 2-3 times, so that the spot is rich in the mixture.
3. Suspend the filter paper vertically in a glass jar containing the solvent (eluent) with the help of a glass rod in such a way that the pencil line (and the spot) remains about 3cm above the solvent level (50% alcohol + distilled water).

OBSERVATIONS AND CALCULATIONS

Substance	Distance travelled by different compounds	Distance travelled by solvent	R _f value
Red ink + Blue ink	A cm (Red ink) B cm (Blue ink)	X cm X cm	A/X B/X

1. Cover the fox and keep it undisturbed. Once the solvent along with the mixture of red and blue inks has risen about 1 cm you can make two different spots of blue and red colour on the filter paper.

2. Take the filter paper out of the fox and mark the distance that the solvent has risen on the paper with a pencil. This is called the solvent front.

3. Dry the paper. Put pencil marks in the centre of the blue & red spots.

4. Measure the distance of the two spots from the original line and the distance of the solvent from the original line.

5. Calculate the R_f values of the blue and red inks by using the formula:

$$R_f = \frac{\text{Distance travelled by the blue or red ink from the point of application}}{\text{Distance travelled by the solvent from the original line}}$$

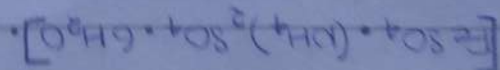
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EXPERIMENT-2.1

Preparation of Inorganic compound

AIM: To prepare a pure sample of ferrous ammonium

sulphate (Mohr's salt)



THEORY

Mohr's salt is prepared by dissolving an equimolar mixture of hydrated ferrous sulphate and ammonium sulphate in water containing a little of sulphuric acid, and then subjecting the resulting solution to crystallisation. When light green crystals of ferrous ammonium sulphate $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ separate out.



Mohr's salt

892

REQUIREMENTS

Two beakers (250ml), china-dish, funnel, funnel stand, glass rod, vacuum-bottle, tripod stand and wire-gauze.

Ferrous sulphate crystals, ammonium sulphate crystals, dilute sulphuric acid and ethyl alcohol.

PROCEDURE

1. Take a 250 ml beaker and wash it with water. Transfer 10g ferrous sulphate and 3.5g ammonium sulphate crystals to it. Add about 2-3 ml of dilute sulphuric acid to prevent the hydrolysis of ferrous sulphate.
2. In another beaker boil about 20ml of water for about 5 minutes to expel dissolved air.
3. Add the boiling hot water to the contents in the first beaker in small installments at a time. Stir with a glass rod until the salts have completely dissolved.
4. Filter the solution to remove undissolved impurities. 4 Transfer the filtrate to a china-dish.
5. Heat the solution in the china-dish for some time to concentrate it to the crystallisation point.
6. Place the china-dish containing saturated solution over a beaker full of cold water. On cooling crystals of Mohr's salt separate out.
7. Decant off the mother liquor quickly. Wash the crystals in the china-dish with a small quantity of alcohol to remove any sulphuric acid sticking to the crystals.

OBSERVATIONS

weight of crystal obtained = g

Expected yield = g

Colour of the crystal =

Shape of the crystal =

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8. Dry the crystals by placing them between filter paper pads.

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EXPERIMENT-2.2

Preparation of Organic compound

Aim: To prepare a pure sample of potassium alum (Fikser),
 $[K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O]$.

THEORY

Potash alum is prepared by dissolving an equimolar mixture of hydrated aluminium sulphate and potassium sulphate in minimum amount of water containing a little of sulphuric acid and then subjecting the resulting solution to crystallising when octahedral crystals of potassium alum separate out.



Potassium	Aluminium	Potash alum
Sulphate	Sulphate	
174	666	

REQUIREMENTS

Two beakers (250ml), china-dish, funnel, funnel stand, glass rod, wash bottle, tripod stand and wire-gauze.

Potassium sulphate, aluminium sulphate and dil. sulphuric acid.

OBSERVATIONS

Weight of crystal obtained = g

Expected yield = g

Colour of the crystal =

Shape of the crystal =

PROCEDURE

1. Take a 250 ml beaker. Wash it with water and then transfer 2.5g potassium sulphate crystals to it. Add about 20 ml of water. Mix to dissolve the crystals warm if required.
2. Take the other 250 ml beaker, wash it with water and then transfer 10g aluminium sulphate crystals to it. Add about 20 ml of water and 1 ml of dilute sulphuric acid to prevent hydrolysis of aluminium sulphate. Heat for about 5 minutes. If milkiness still persists, filter the solution.
3. Mix the two solutions in a china-dish and place the china-dish on a wire-gauze placed over a burner. Stir the solution with a glass rod. Concentrate the solution till the crystallisation point is reached. Place the dish over a beaker containing cold water.
4. Spoon the crystals of potassium separate out. Decant off the mother liquor and wash the crystals with a small quantity of ice-cold water.
5. Dry the crystals by placing them between filter paper pads or by spreading them over porous plate.

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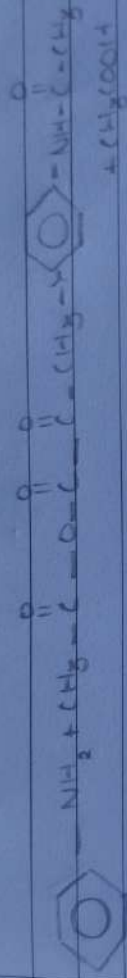
EXPERIMENT-3

Preparation of organic compound

Aim: To prepare acetanilide from aniline

THEORY

Acetanilide is prepared by acetylating aniline with acetic anhydride in the presence of glacial acetic acid. The chemical equation can be written as:



APPARATUS

Round bottom flask (100 ml), water condenser, wire-gauge, glass rod, tripod stand, burner, iron-stand, clamp, measuring cylinder, etc.

CHEMICALS REQUIRED

Aniline = 5 ml

Acetic anhydride = 5 ml

Glacial acetic acid = 5 ml

PROCEDURE

1. Take 5ml of acetic anhydride in a clean dry round bottom flask and add 5ml of glacial acetic acid and shake the contents thoroughly.
2. To this mixture taken in the flask, add 5ml of aniline and fit a water condenser.
3. Place the flask on a wire-gauge placed on a tripod stand.
4. Boil the mixture for 10-15 minutes.
5. Allow the mixture to cool. Detach the condenser and pour the liquid into about 50ml ice-cold water contained in a beaker. During addition, stir vigorously the contents of the beaker with the help of glass rod.
6. Filter the white precipitate which separate out and wash with cold water.
7. Recrystallise from hot water containing a few drops of ethyl alcohol. Weigh the crystals and record the yield.
8. Determine the melting point of the compound.

RESULT

Weight of acetanilide obtained =g

Melting point of acetanilide =°C

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